

Solution Sheet-04

1. **Problem:** Given $n^a P(|X| > n) \rightarrow 0$ as $n \rightarrow \infty$, prove $E[|X|^p] < \infty$ for $0 < p < a$.

Solution:

We use the integral formula for the expectation of a non-negative random variable $Y = |X|^p$:

$$E[|X|^p] = \int_0^\infty P(|X|^p > t) dt = \int_0^\infty P(|X| > t^{1/p}) dt.$$

Let $u = t^{1/p}$, so $t = u^p$ and $dt = pu^{p-1} du$.

$$E[|X|^p] = \int_0^\infty pu^{p-1} P(|X| > u) du.$$

Split the integral into $\int_0^C + \int_C^\infty$. The first part is finite (probability ≤ 1). For the second part, we use the given condition. Since $\lim_{n \rightarrow \infty} n^a P(|X| > n) = 0$, there exists a constant $C > 0$ such that for all $u > C$:

$$u^a P(|X| > u) \leq 1 \implies P(|X| > u) \leq \frac{1}{u^a}.$$

Substituting this into the integral:

$$\int_C^\infty pu^{p-1} P(|X| > u) du \leq \int_C^\infty pu^{p-1} \cdot u^{-a} du = \int_C^\infty pu^{p-1-a} du.$$

This integral converges if the power $p-1-a < -1$, which simplifies to $p < a$. Since $0 < p < a$, the integral converges, proving $E[|X|^p] < \infty$.

2. **Problem:** Prove $P(X \geq x) \leq \frac{\sigma^2}{\sigma^2 + x^2}$ for $x > 0$.

Solution:

Let $x > 0$. For any $c > 0$, the event $\{X \geq x\}$ is the same as $\{X + c \geq x + c\}$. Since $X + c$ is essentially squared in Markov's inequality, we consider:

$$P(X \geq x) = P(X + c \geq x + c) \leq P((X + c)^2 \geq (x + c)^2).$$

By Markov's inequality:

$$P((X + c)^2 \geq (x + c)^2) \leq \frac{E[(X + c)^2]}{(x + c)^2}.$$

$$E[(X + c)^2] = E[X^2 + 2cX + c^2] = \text{Var}(X) + (E[X])^2 + 2cE[X] + c^2.$$

Since $E[X] = 0$ and $\text{Var}(X) = \sigma^2$:

$$E[(X + c)^2] = \sigma^2 + c^2.$$

So, $P(X \geq x) \leq \frac{\sigma^2 + c^2}{(x + c)^2}$. To minimize the upper bound, we differentiate with respect to c and set to 0. This yields $c = \sigma^2/x$. Substituting $c = \sigma^2/x$:

$$\frac{\sigma^2 + (\sigma^2/x)^2}{(x + \sigma^2/x)^2} = \frac{\sigma^2(1 + \sigma^2/x^2)}{x^2(1 + \sigma^2/x^2)^2} = \frac{\sigma^2}{x^2(1 + \sigma^2/x^2)} = \frac{\sigma^2}{x^2 + \sigma^2}.$$

(The case for $x < 0$ follows similarly or by applying the result to $-X$).

3. Solution:

We apply the inequality derived in Q2 to the random variable $Y = X^2$. We want to bound $P(|X| > K\sigma) = P(X^2 > K^2\sigma^2)$. Let $Y = X^2$. Then $E[Y] = \sigma^2$ and $E[Y^2] = E[X^4] = \mu_4$. $\text{Var}(Y) = \mu_4 - (\sigma^2)^2 = \mu_4 - \sigma^4$. We want $P(Y > K^2\sigma^2)$. Let threshold $t = K^2\sigma^2$. Note that $t > E[Y]$ since $K > 1$. Using the one-sided Chebyshev inequality for Y :

$$P(Y \geq t) \leq \frac{\text{Var}(Y)}{\text{Var}(Y) + (t - E[Y])^2}.$$

Substitute parameters:

$$P(X^2 \geq K^2\sigma^2) \leq \frac{\mu_4 - \sigma^4}{(\mu_4 - \sigma^4) + (K^2\sigma^2 - \sigma^2)^2}.$$

Denominator simplification: $\mu_4 - \sigma^4 + \sigma^4(K^2 - 1)^2 = \mu_4 - \sigma^4 + \sigma^4(K^4 - 2K^2 + 1) = \mu_4 + \sigma^4K^4 - 2\sigma^4K^2$. Thus:

$$P(|X| > K\sigma) \leq \frac{\mu_4 - \sigma^4}{\mu_4 + K^4\sigma^4 - 2K^2\sigma^4}.$$

4. Solution:

The PDF corresponds to a Gamma distribution with shape $\alpha = A + 1$ and rate $\beta = 1$. Mean $\mu = \frac{\alpha}{\beta} = A + 1$. Variance $\sigma^2 = \frac{\alpha}{\beta^2} = A + 1$. We want to show $P(0 < X < 2(A + 1)) > \frac{A}{A+1}$. The interval is $(0, 2\mu)$, which corresponds to $|X - \mu| < \mu$. Note that $\mu = A + 1$. The inequality is $|X - (A + 1)| < A + 1$. In terms of standard deviations: $\mu = \sqrt{A + 1} \cdot \sigma$. So we are looking for $P(|X - \mu| < \sqrt{A + 1} \cdot \sigma)$. By Chebyshev's Inequality:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

Here $k = \sqrt{A + 1}$.

$$P(|X - (A + 1)| < A + 1) \geq 1 - \frac{1}{A + 1} = \frac{A}{A + 1}.$$

Since X is non-negative, $|X - (A + 1)| < A + 1 \iff -(A + 1) < X - (A + 1) < A + 1 \iff 0 < X < 2(A + 1)$. Thus, $P(0 < X < 2(A + 1)) > \frac{A}{A+1}$.

5. Solution:

Consider a random variable $X \sim \text{Exponential}(1)$. PDF $f(x) = e^{-x}$ for $x > 0$. Mean $\mu = 1$, Variance $\sigma^2 = 1$. Consider the event $X \geq k + 1$.

$$P(X \geq k + 1) = \int_{k+1}^{\infty} e^{-x} dx = e^{-(k+1)}.$$

We can view this event as a deviation from the mean: $X - 1 \geq k$. Since $X - 1 \geq k$ implies $|X - 1| \geq k$, we have:

$$P(X - 1 \geq k) \leq P(|X - 1| \geq k).$$

By Chebyshev's inequality for X :

$$P(|X - 1| \geq k) \leq \frac{\sigma^2}{k^2} = \frac{1}{k^2}.$$

Combining these:

$$e^{-(k+1)} \leq \frac{1}{k^2} \implies k^2 \leq e^{k+1}.$$

6. Solution:

For any $t > 0$, consider the event $\{X > 0\}$. This is equivalent to $\{e^{tX} > 1\}$ (since exponential is strictly increasing). By Markov's Inequality applied to the non-negative random variable $Y = e^{tX}$:

$$P(e^{tX} > 1) \leq \frac{E[e^{tX}]}{1} = \varphi(t).$$

Since this holds for *any* $t > 0$, it must hold for the infimum:

$$P(X > 0) \leq \inf_{t>0} \varphi(t).$$

Also, probabilities are always ≤ 1 . Thus the result holds.

7. Solution:

To find $f_X(x)$, integrate joint PDF over $y \in (-\infty, \infty)$:

$$f_X(x) = \int_{-\infty}^{\infty} \frac{c}{2\pi(c^2 + x^2 + y^2)^{3/2}} dy.$$

Let $A = c^2 + x^2$. We need to evaluate $\int_{-\infty}^{\infty} \frac{c}{2\pi(A+y^2)^{3/2}} dy$. Substitute $y = \sqrt{A} \tan \theta$, $dy = \sqrt{A} \sec^2 \theta d\theta$.

$$(A + y^2)^{3/2} = (A(1 + \tan^2 \theta))^{3/2} = A^{3/2} \sec^3 \theta.$$

$$\int \frac{1}{(A + y^2)^{3/2}} dy = \int \frac{\sqrt{A} \sec^2 \theta}{A^{3/2} \sec^3 \theta} d\theta = \frac{1}{A} \int \cos \theta d\theta.$$

Limits: $y \rightarrow \pm\infty \implies \theta \rightarrow \pm\pi/2$. Integral value = $\frac{1}{A} [\sin \theta]_{-\pi/2}^{\pi/2} = \frac{2}{A}$. Substituting back:

$$f_X(x) = \frac{c}{2\pi} \cdot \frac{2}{c^2 + x^2} = \frac{c}{\pi(c^2 + x^2)}.$$

Similarly, $f_Y(y) = \frac{c}{\pi(c^2 + y^2)}$. (These are Cauchy distributions).

8. Solution:

Let $S = 1 - p_1 - p_2$. The joint PMF is:

$$P(x, y) = \frac{(x + y + k - 1)!}{x!y!(k - 1)!} p_1^x p_2^y S^k.$$

To find marginal $P(X = x)$, sum over $y = 0$ to ∞ :

$$P(X = x) = \frac{p_1^x S^k}{x!(k - 1)!} \sum_{y=0}^{\infty} \frac{(x + y + k - 1)!}{y!} p_2^y.$$

Rearrange the factorial term to recognize the Negative Binomial series expansion:

$$\frac{(x + y + k - 1)!}{y!} = (x + k - 1)! \cdot \binom{x + y + k - 1}{y}.$$

$$P(X = x) = \frac{p_1^x S^k (x + k - 1)!}{x!(k - 1)!} \sum_{y=0}^{\infty} \binom{x + y + k - 1}{y} p_2^y.$$

Using the identity $\sum_{j=0}^{\infty} \binom{j+r-1}{j} z^j = (1-z)^{-r}$: Here $r = x+k$ and $z = p_2$.

$$\sum_{y=0}^{\infty} \binom{y+x+k-1}{y} p_2^y = (1-p_2)^{-(x+k)}.$$

Substitute back:

$$P(X=x) = \binom{x+k-1}{x} p_1^x S^k (1-p_2)^{-(x+k)}.$$

Note $S = 1 - p_1 - p_2$, so $\frac{S}{1-p_2} = \frac{1-p_1-p_2}{1-p_2} = 1 - \frac{p_1}{1-p_2}$. Let $P_1 = \frac{p_1}{1-p_2}$. Then:

$$P(X=x) = \binom{x+k-1}{x} \left(\frac{p_1}{1-p_2} \right)^x \left(\frac{S}{1-p_2} \right)^k.$$

This is a Negative Binomial distribution $NB(k, P_1)$. Similarly for Y .

9. Solution:

Region: $0 < x < y < z < u < \infty$. Integral: $\int_0^\infty \int_0^u \int_0^z \int_0^y e^{-u} dx dy dz du$. Inner integrals: $\int_0^y dx = y$. $\int_0^z y dy = z^2/2$. $\int_0^u z^2/2 dz = u^3/6$. Final integral: $\int_0^\infty \frac{u^3}{6} e^{-u} du = \frac{1}{6} \Gamma(4) = \frac{3!}{6} = 1$. Yes, it is a valid PDF.

Probability $P(X < 7)$: This requires finding the marginal PDF of X . $f_X(x) = \int_x^\infty \int_y^\infty \int_z^\infty e^{-u} du dz dy$. $\int_z^\infty e^{-u} du = e^{-z}$. $\int_y^\infty e^{-z} dz = e^{-y}$. $\int_x^\infty e^{-y} dy = e^{-x}$. So $X \sim \text{Exp}(1)$. $P(X < 7) = 1 - e^{-7}$.

10. Solution:

1. Verification of Joint PDF: To be a valid PDF, the integral over the support must equal

1. Let $S = X + Y + Z + U$. The volume element in the simplex domain $\sum x_i = s$ scales with s^{n-1} . Specifically, for a symmetric function $g(\sum x_i)$, the PDF of the sum S is given by:

$$f_S(s) = \frac{s^{n-1}}{(n-1)!} \times (\text{normalization from } f).$$

Here $n = 4$ and the kernel is $\frac{24}{(1+s)^5}$.

$$f_S(s) = \frac{s^3}{3!} \cdot \frac{24}{(1+s)^5} = \frac{24}{6} \frac{s^3}{(1+s)^5} = \frac{4s^3}{(1+s)^5}, \quad s > 0.$$

We verify $\int_0^\infty f_S(s) ds = 1$. Let $I = \int_0^\infty 4s^3(1+s)^{-5} ds$. This is related to the Beta function $B(x, y) = \int_0^\infty \frac{t^{x-1}}{(1+t)^{x+y}} dt$. Here $x = 4$ and $x + y = 5 \implies y = 1$.

$$I = 4 \cdot B(4, 1) = 4 \cdot \frac{\Gamma(4)\Gamma(1)}{\Gamma(5)} = 4 \cdot \frac{3! \cdot 0!}{4!} = 4 \cdot \frac{6}{24} = 1.$$

Since the integral is 1 and $f \geq 0$, it is a valid PDF.

(a) Find $\mathbb{P}(X > Y > Z > U)$: The joint PDF $f(x, y, z, u)$ is symmetric with respect to the variables x, y, z, u (it depends only on their sum). The variables are exchangeable. There are $4! = 24$ possible strict orderings of the four random variables (e.g., $X > Y > Z > U$, $Y > X > Z > U$, etc.). Due to symmetry, each ordering is equally likely.

$$\mathbb{P}(X > Y > Z > U) = \frac{1}{4!} = \frac{1}{24}.$$

(b) **Find** $\mathbb{P}(X + Y + Z + U > 1)$: Let $S = X + Y + Z + U$. We found the PDF of S in step 1:

$$f_S(s) = \frac{4s^3}{(1+s)^5}, \quad s > 0.$$

We need to calculate $\mathbb{P}(S > 1) = \int_1^\infty \frac{4s^3}{(1+s)^5} ds$. Substitute $t = 1 + s$, so $s = t - 1$ and $ds = dt$. Limits change from $(1, \infty)$ to $(2, \infty)$.

$$\begin{aligned} \mathbb{P}(S > 1) &= 4 \int_2^\infty \frac{(t-1)^3}{t^5} dt = 4 \int_2^\infty \frac{t^3 - 3t^2 + 3t - 1}{t^5} dt. \\ &= 4 \int_2^\infty (t^{-2} - 3t^{-3} + 3t^{-4} - t^{-5}) dt. \end{aligned}$$

Integrating term by term:

$$\begin{aligned} &= 4 \left[\frac{t^{-1}}{-1} - 3 \frac{t^{-2}}{-2} + 3 \frac{t^{-3}}{-3} - \frac{t^{-4}}{-4} \right]_2^\infty \\ &= 4 \left[-\frac{1}{t} + \frac{3}{2t^2} - \frac{1}{t^3} + \frac{1}{4t^4} \right]_2^\infty. \end{aligned}$$

At the upper limit (∞), all terms vanish. At the lower limit (2):

$$\begin{aligned} &= 4 \left(0 - \left[-\frac{1}{2} + \frac{3}{2(4)} - \frac{1}{8} + \frac{1}{4(16)} \right] \right) \\ &= 4 \left(\frac{1}{2} - \frac{3}{8} + \frac{1}{8} - \frac{1}{64} \right) \\ &= 4 \left(\frac{32 - 24 + 8 - 1}{64} \right) = 4 \left(\frac{15}{64} \right) = \frac{15}{16}. \end{aligned}$$

Answer: $\mathbb{P}(X + Y + Z + U > 1) = \frac{15}{16}$.

11. **Solution:**

$P(X = Y) = \iint_D f(x, y) dx dy$ where $D = \{(x, y) : x = y\}$. The region D is a line in the xy -plane. The area (2D Lebesgue measure) of a line is 0. Since $f(x, y)$ is continuous (and bounded on bounded domains), the integral over a set of measure zero is 0. **Geometric Explanation:** For continuous variables, probability corresponds to volume under the surface $f(x, y)$. A line has no width, so the volume of the slice sitting above the line is zero. For discrete variables, probability mass is concentrated at points. If mass points lie on the line $x = y$, the sum can be positive.

12. **Solution:**

(a) **Constant C:** Volume must be 1. Area of unit disk is $\pi(1)^2 = \pi$. Thus $C = \frac{1}{\pi}$.

(b) **Marginal $f_X(x)$:** For a fixed $x \in (-1, 1)$, y ranges from $-\sqrt{1-x^2}$ to $\sqrt{1-x^2}$.

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{1}{\pi} [y]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} = \frac{2\sqrt{1-x^2}}{\pi}, \quad -1 < x < 1.$$

(c) **Independence:** X and Y are **not** independent. The joint PDF is constant, but the domain is not a rectangle product space. Also, check product of marginals: $f_X(x)f_Y(y) = \frac{4}{\pi^2} \sqrt{1-x^2} \sqrt{1-y^2} \neq \frac{1}{\pi}$.

13. Solution:

$f(x, y) = 4xy$ on unit square. $P(Y/X > 1) = P(Y > X)$. Integrate $f(x, y)$ over region $0 < x < 1$ and $x < y < 1$.

$$P(Y > X) = \int_0^1 \int_x^1 4xy \, dy \, dx.$$

Inner integral: $\int_x^1 4xy \, dy = 2x[y^2]_x^1 = 2x(1 - x^2) = 2x - 2x^3$. Outer integral: $\int_0^1 (2x - 2x^3) \, dx = [x^2 - \frac{1}{2}x^4]_0^1 = 1 - 0.5 = 0.5$.